

Multiscale cortical parcellation based on geodesic distance and hierarchical clustering

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Abstract—Brain neuronal networks of structural and functional connections have a hierarchical organization and a complex relationship between them. To study brain dynamics, it is important to identify the cortical level of parcellation of greater metastability. This paper presents a new multiscale cortical parcellation method based on the geodesic distance between vertices of the cortical surface and agglomerative hierarchical clustering, starting from an anatomical parcellation. First, the centroids of each region are efficiently calculated using the geodesic distance between the region's vertices. Then, an affinity graph is constructed between the region centroids, based on the geodesic distance, from which a dendrogram is constructed using hierarchical clustering. Finally, an adaptive tree partitioning method is employed to obtain parcellations at various granularity levels, producing a multiscale parcellation. Furthermore, we propose an optimized method for the calculation of structural connectomes for each parcellation level. This framework will be made available and can be applied to different fine-grained parcellations. Additional information, such as structural connectivity information can be easily added to the framework. In future work this multiscale cortical parcellation will allow for simulations of cerebral dynamics at different levels.

Index Terms—multiscale cortical parcellation, geodesic distance, hierarchical clustering, dendrogram

I. INTRODUCTION

The brain can be analyzed as a complex network of interacting regions, often referred to as the human connectome (HC), which integrates and segregates regions depending on the task to be performed [1]. The connectome is composed of two main elements. First, gray matter (GM), composed mainly of neuronal bodies, located in the cortical and sub-cortical regions, is responsible for information processing and response to stimuli. Second, white matter (WM), composed of myelinated axons, is responsible for interconnecting different regions of gray matter [2]. Non-invasive in vivo Magnetic

Resonance Imaging (MRI) techniques allow us to obtain a representation of the connections between different brain areas. These techniques include diffusion-weighted magnetic resonance imaging (dMRI) [3], [4] and resting-state functional MRI (rfMRI) [5].

To obtain and study the human connectome, a crucial step is to define the gray matter parcellation, over which the connectivity profiles are calculated. Parcellation can be performed on GM volumes or GM surfaces [6]. However, the use of cortical surfaces has several advantages, such as the preservation of curvature, a higher resolution than volumes, and the enhancement of neuroimaging processes such as the registration and computation of cortical thickness [6]–[8]. The analysis derived from parcellation enables the study of different pathological conditions, development, and aging.

In the literature, several methods have been described to obtain representative parcellations of a population. For example, methods based on structural connectivity, that can include the use of fascicle atlases or averaging individual structural connections [9]–[12]. Other methods are based on resting-state fMRI [13], or multiple cortical features such as cortical architecture, function, connectivity, and topography [14].

Some studies have focused on obtaining different levels or scales of parcellation, ranging from fine- to coarse-grained parcellations, based on a base parcellation [15]–[17]. The analysis stemming from these multiple levels of parcellation sheds light on a matter that has been a subject of neuroscience research for years: the function-anatomical organization of the brain. In [15], a hierarchical clustering method based on tractography is proposed, which encodes information about the underlying structure at all levels of granularity in a dendrogram. Depending on the number of fibers that connect a pair of regions, different levels of cortical parcellation are achieved. However, the method only used horizontal cuts and had to perform a complex tree matching between subjects. In [16], a method creates group-wise connectivity-based parcellations

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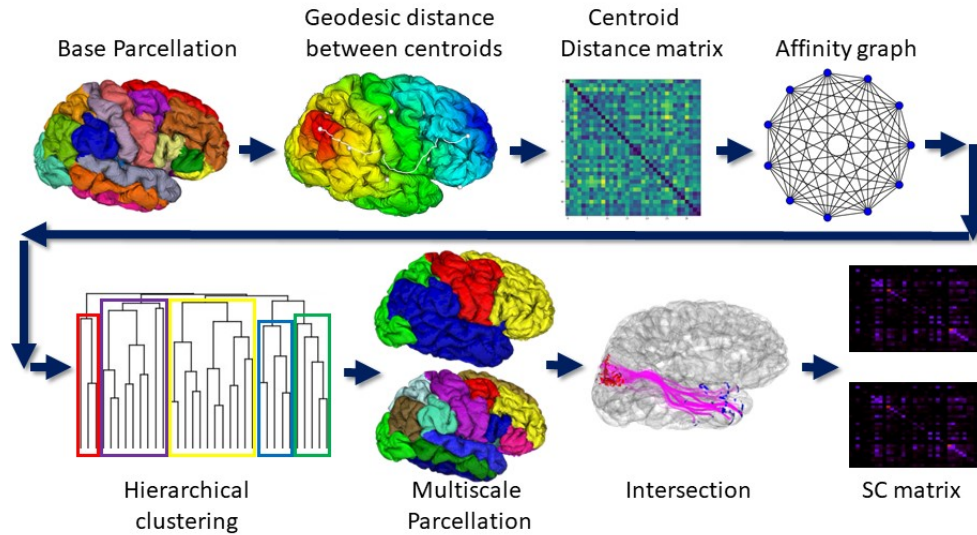


Fig. 1. Proposed method to obtain the multiscale cortical parcellation and the SC matrix for each level

using spatial information from an average sub-parcel overlap matrix and a mean connectivity matrix. However, parcellations are not defined on a multiscale structure.

In [17], hierarchical clustering is applied to a base parcellation based on rfMRI of 2514 regions to obtain different parcellation levels. To achieve this, the authors used a mean structural connectivity matrix (SC) and a mean functional connectivity matrix (FC) of 12 healthy subjects. They then analyze the dendrogram resulting from the FC matrix and find the maximum values of Newman's modularity (Q) at levels ranging from 12 to 25 regions, with the 20-region level providing a remarkably large value of modularity Q for the SC matrix. In other words, a partition inferred exclusively from functional measurements seems to lead to an ordering of the structural/wiring data. This interesting work sheds light on the relevance of the study of brain connectivity at different levels of granularity. However, this method uses a volume gray matter parcellation with a complex subdivision, which may difficult the calculation of tractography-based analyses and does not provide compact regions at all levels.

Hence, we propose a framework for the generation of multiscale cortical parcellations over a surface representation of the cortex. The method is based on hierarchical clustering and geodesic distance, and uses a tree partitioning method, depending on the maximum distance between regions in a cluster, to obtain different parcellation scales. An easy integration of connectivity information is possible by modifying the mean distance matrix, as in [16]. The method establishes the correspondence between all levels of parcellation, and efficiently calculates the subject's connectome for each level. For testing, it was applied to the Desikan-Killiany anatomical atlas (34 regions per hemisphere) as a base parcellation.

II. MATERIALS AND METHODS

A. Desikan-Killiany atlas, tractography and cortical mesh

We used data from a random subject of the ARCHI database [18], which contains anatomical MRI and HARDI data with special acquisition sequences from a 3T MRI scanner with a 12-channel head coil (Siemens, Erlangen). The MRI protocol included the acquisition of a T1 image dataset using a MPRAGE sequence (160 slices, TE/TR = 2.98/2300 ms, matrix = 256×240 ; voxel size = $1 \times 1 \times 1.1$ mm), a single shell HARDI SS-EPI dataset along 60 optimized diffusion weighted directions with a $b = 1,500$ s/mm², and a field map B0, (70 slices, TE = 93 ms, TR = 14,000 ms, matrix = 128×128). To obtain the tractography data were pre-processed with BrainVISA/Connectomist-2.0 software. The main artifacts, such as noise, susceptibility effects, geometric distortions, and eddy currents were corrected. Then, the analytic Q-ball model [19] was computed. Whole-brain streamline deterministic tractography was calculated, with one seed per voxel at T1 resolution, a tracking step of 0.2 mm and a maximum curvature angle of 30°. Resulting tractography contain about one million fibers. The cortical mesh of the subject (81920 polygons and 40962 vertices per hemisphere) and the labels associated with each cortical mesh vertex, corresponding to the Desikan-Killiany atlas (34 regions per hemisphere), were calculated from the T1 image using Freesurfer and BrainVisa. These cortical meshes present anatomical correspondence between subjects.

B. Parcellation method

Fig. 1 illustrates the proposed method, which is applied separately to each hemisphere. First, the geodesic distance is calculated for each region in the base parcellation. Next, a distance matrix is calculated based on the geodesic distance between the centroids of each pair of regions. These values are entered into an affinity graph, and an agglomerative

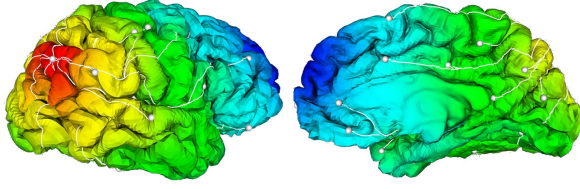


Fig. 2. Geodesic distance paths from one centroid to the rest of the centroids.

hierarchical clustering is applied to obtain a dendrogram. A tree partitioning method, depending on the maximum distance between regions in a cluster, was used to obtain different parcellation scales. The method establishes and saves the correspondence between all levels of parcellation. Finally, it efficiently calculates the subject's connectome for each level, based on a whole-brain tractography dataset.

1) *Geodesic distance and region centroids*: The first step involves calculating the centroid of each region of the base (fine-grained) parcellation. On the labeled cortical mesh, the geodesic distance is calculated between the vertices of each region of interest (ROI). The geodesic distance differs from the Euclidean distance, as it is not calculated by a straight line between two points in the 3D space, but over the surface. It is measured between two points that belong to the surface and corresponds to the length of the shortest possible path between them, always following the surface. The mesh vertex closest to all the other region's vertices is selected as the centroid of the region. Based on [20], the method first creates a graph in which the nodes represent all the vertices of the mesh region and the edge weights are the Euclidean distances between them. To determine the geodesic distance between a pair of vertices, the method selects the shortest path from one vertex to another on the graph.

2) *Distance matrix*: To obtain the distance matrix, the geodesic distance between the centroids of each pair of ROIs is used as the metric. Fig. 2 shows the geodesic distance paths from one centroid to the rest of the centroids on the mesh. A symmetric matrix is obtained, with dimensions 34x34 in the case of the Desikan-Killiany atlas. Since the meshes are closed surfaces per hemisphere, we performed all the analyzes separately for each hemisphere.

3) *Affinity graph*: Affinity measures are commonly employed in clustering literature, specifically using a Gaussian Kernel. In addition, our hierarchical clustering algorithm is optimized to work with graphs. Hence, the data from the matrix is entered into an affinity graph, whose weights are defined by Equation 1, where d_{ij} represents the distance between elements i and j , and σ^2 is the parameter that defines the scale of similarity ($\sigma = 60mm$) [16], [21].

$$a_{ij} = e^{-d_{ij}/\sigma^2} \quad (1)$$

4) *Hierarchical clustering*: Then, a dendrogram is constructed using an average-link agglomerative hierarchical clustering algorithm on the affinity graph. The leaves represent the

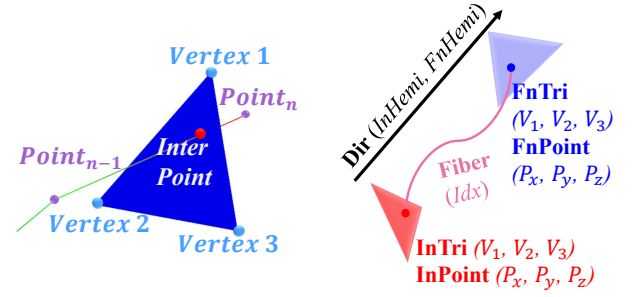


Fig. 3. The left side of the figure illustrates a representation of the intersection of one fiber extremity and one triangle of the cortical surface. The right side represents the returned values of the algorithm for one fiber that intersects with both extremities.

parcels in the base parcellation, and in each iteration, the two closest parcels are merged to form a node, finally resulting in a hierarchical tree.

5) *Adaptative partition of the dendrogram*: To achieve a multiscale parcellation, we implemented an adaptive partition method using the hierarchical tree, based on [21]. It uses a maximum distance (Dm) between the leaves of a cluster, in this case composed of the centroids of the regions. Starting from the top node, the algorithm calculates the maximum pairwise distance between all its descendant leaves. If the distance is smaller than or equal to Dm , the leaves are grouped into a cluster. On the other hand, if the distance exceeds Dm , the child nodes are added to a queue for further analysis using the same criterion as their parent. This approach results in a parcellation with varying numbers of parcels and parcel sizes for each Dm used. Some regions do not merge or separate from the rest to create new parcels at the subsequent level, depending on the selected Dm for forming that parcellation's level.

As a result, the algorithm obtains and stores the multiscale parcellation, that is, the regions associated with each cluster (or parcel) for each level of parcellation, and the descendants of the clusters of the higher level, using the parent-child relationship inherent to the dendrogram.

6) *Calculation of structural connectome*: We developed an algorithm to calculate the structural connectome, based on a whole-brain tractography dataset, and the meshes of both hemispheres with labels corresponding to any parcellation level. For that, the intersection of fibers and cortical surfaces requires to be calculated in an efficient manner. Inspired by previous studies [22], [23], we implemented an algorithm capable of determining the intersection of a whole-brain tractogram and, as an intermediate process, distinguishing inter- and intra-hemispheric fibers.

The algorithm for intersecting a fiber and the cortical surfaces is based on the Möller-trumbore ray-triangle intersection [24], which determines whether a ray intersects a triangle. In our case, the ray is represented as a vector composed of the two extreme points of the fiber, and the triangle is represented by the three 3D cortical mesh vertices that compose it (see Fig. 3). Our implementation returns a

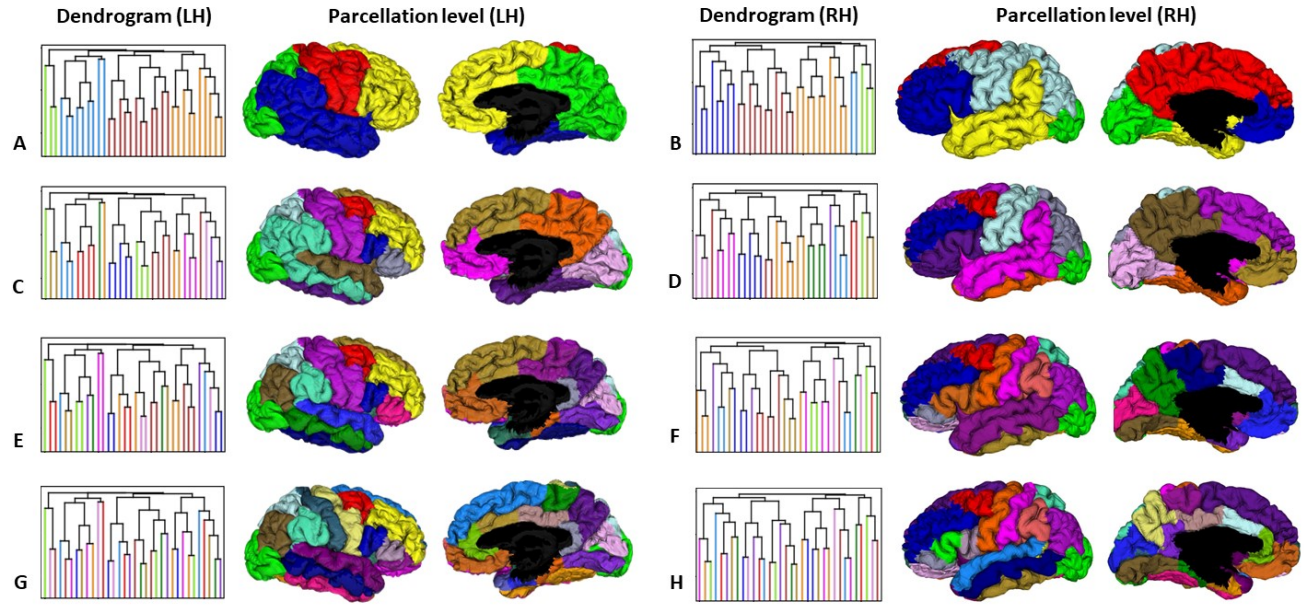


Fig. 4. Four levels of parcellation, depending on the selected Dm (in mm). Level1, $Dm=180$, (A) 4 parcels, (B) 5 parcels. Level 2, $Dm=84$, (C) 14 parcels, (D) 13 parcels. Level 3, $Dm=42$, (E) 24 parcels, (F) 24 parcels. Level 4, $Dm=10$, (G) 34 parcels, (H) 34 parcels

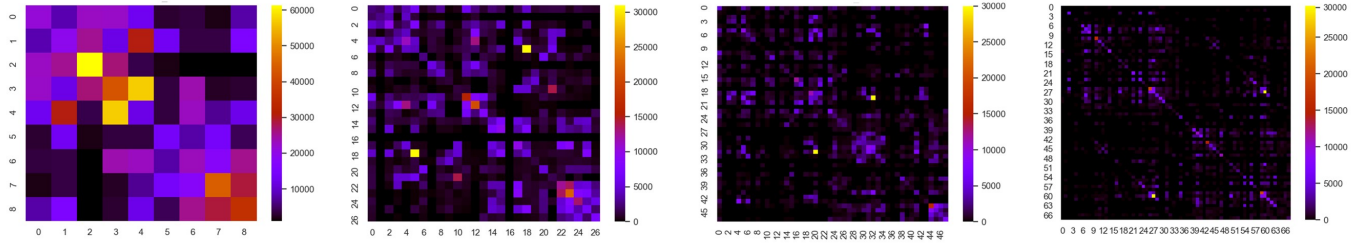


Fig. 5. Connectivity matrices for the entire brain, corresponding to the four levels described, levels 1 to 4 from left to right.

boolean indicating if the ray intersects the triangle, and the 3D intersection point over the triangle. This is an iterative algorithm implemented in C++, which at each iteration verifies the existence of an intersection of the two ends of each fiber and a triangle of the cortical surface. This process has a high computational cost, therefore, some optimizations were performed to reduce the execution time. These include an OpenMP parallelization and a division of the 3D space into cubic cells to reduce the search time for the intersection of a fiber with a triangle of the mesh. The algorithm also calculates a projection of fibers one point backward and two forward, to ensure that there are no missing valid intersections in the periphery of the cubic cells. The algorithm generates a file with information associated with the intersecting fibers, such as the index of the fiber, the hemisphere intersected by each fiber end, the initial and final points of intersection, and finally, the initial and final intersected triangles (see Fig. 3). With the output information, we implemented a function to compute the structural connectome of each parcellation, integrating the labeling information associated with each parcellation level.

Note that the intersection must be calculated only once for the entire multiscale parcellation. Furthermore, Using the result or intermediate calculations of the intersection algorithm, some processes could be applied in order to obtain more information about structural connectivity, such as the differentiation of inter- and intra-hemispheric fibers, that could be used to obtain SC information to add to our parcellation framework.

III. RESULTS

We implemented the intersection method in C++ and the other methods in Python programming language version 3.9. The entire algorithm was executed on a computer with a 6-core Intel Core i5-8600K CPU 3.60 GHz, 9MB of shared L3 cache, 16 GB of RAM, using Ubuntu 20.04.

Centroid Distance matrices for both hemispheres were computed in 23.21 minutes. Next, we obtained the affinity graphs, which took 0.81 seconds. The implementation of the hierarchical clustering method and the adaptive partition of the trees to form the four parcellation levels took 8.35 seconds.

The results in Fig. 4 display four scales of parcellation. As we move from a lower level to a higher one, parcels

with a centroid distance smaller than the chosen threshold for that level, merge to form a cluster while maintaining the hierarchical organization across all levels.

The intersection between fibers and cortical surfaces was achieved in 27 seconds. On the other hand, the execution time to determine the structural connectome is dependent on the number of parcels, its distribution and size. Nevertheless, the maximum execution time was 6 seconds in the fourth level of parcellations. The results for the connectome in the four proposed levels are illustrated in Fig. 5.

IV. DISCUSSION AND CONCLUSIONS

We propose an efficient framework to create a multiscale cortical parcellation using geodesic distance and hierarchical clustering, with configurable parameters and based on any base parcellation. The generated parcels exhibit spatial contiguity. This is attributed to the utilization of geodesic distance as a criterion for constructing the distance matrix, however, a stronger constraint could be added to force contiguity between regions of a parcel. With this adaptive partitioning of the hierarchical tree, we can generate as many levels of parcellation as desired, depending on the number of parcels in the base parcellation and the selection of different values of D_m for each level. But, of course, if two values of (D_m) are too close, some parcels could not be subdivided at the next level. Hierarchical information is preserved at all parcellation levels for subsequent analysis.

Note that the algorithm was applied to one subject, but multiple subjects can be easily considered by calculating a mean distance matrix for all subjects, since the correspondence between subjects is given by the cortical meshes. Also, since our base parcellation is symmetrical, we could force symmetrical parcellations at every level by using only one average matrix for both hemispheres. However, for this work, we applied the generic algorithm, which does not assume a symmetrical base parcellation.

This algorithm will be applied to fine-grained parcellations, and information about structural connectivity will be added to the distance matrix to obtain parcels that are spatially contiguous and also based on structural connections. In future work, our multiscale cortical parcellation will allow for simulations of cerebral dynamics at different levels, with the aim of identifying the level with the greatest metastability.

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